Report on -div A grad

Qile Yan

August 18, 2024

1 Problem setting

Let $A : \mathbb{R}^d \to \mathbb{R}^{d\times d}$ be a symmetric and uniformly elliptic field in the sense of

$$e \cdot A(x)e \geq \frac{1}{C_0}|e|^2, \quad |A(x)e| \leq |x|, \quad \text{for all } x, e \in \mathbb{R}^d.$$

In [1], they consider the acoustic operator

$$H_a := -\nabla \cdot A\nabla, \quad \text{on } L^2(\mathbb{R}^d).$$

where A is periodic, quasiperiodic or random.

Some discussions: Recall that for the operator $L = -\Delta + V$, one can define its landscape function u as the solution to $Lu = 1$. The conjugation of L is defined as

$$\tilde{L}g = -\frac{1}{u^2}\mathrm{div}(u^2\nabla g) + \frac{1}{u}g.$$

Then $L\psi = \lambda\psi$ if and only if $\phi := \psi/u$ satisfies $\tilde{L}\phi = \lambda\phi$.

Now, the eigenvalue problem is

$$-\mathrm{div}(A\nabla u) = \lambda u.$$

We may check the eigenfunctions of the following operator

$$-\mathrm{div}(A\nabla u) = \lambda Au.$$

In the numerical experiment, we also test the following

$$-\mathrm{div}(\nabla u) = \lambda Au$$

2 Numerical experiment

One simple way to describe the localization of eigenfunction is the participation ratio of an eigenfunction which is defined as

$$\mathrm{PR(f)} = \frac{1}{|\Omega|}\frac{(\int_\Omega |f|^2)^2}{\int_\Omega f^4}.$$

Note that $0 \leq \mathrm{PR(f)} \leq 1$.

2.1 1d random displacement model in [4]

In Theorem 1.2 of [4], it was proved that when $d = 1$ and

$$A(x) = \frac{1}{1 + \sum_{n \in Z} f(x - n - d_n(\omega))}$$

where $\text{supp}(f) \subset [-s, s], f \in L^1, 1 + f > 0, f \neq 0$ and the displacements $\{d_n(\omega)\}$ are i.i.d random variables taking values in $[-d_{max}, d_{max}]$ with $d_{max} + s < 1/2$ then the operator H_a almost surely has dense pure point spectrum with exponentially decaying eigenfunctions.

The test codes is under the folder:grad_A_div/1d

Consider solving

$$-(Au')' = \lambda u$$

on a closed interval $[x_0, x_1]$. Let $s = 1/4$ and define

$$f(x) = 1/8[\max\{(1-x^2/s^2)^3, 0\}(3x^2+1)]' = \begin{cases} \frac{6}{8}x(1 - x^2/s^2)^3 - \frac{6}{8s^2}x(1 - x^2/s^2)^2(3x^2 + 1) & x \in [-s, s] \\ 0, & x \notin [-s, s] \end{cases}$$

Then $\text{supp}(f) \subset [-s, s], f \in C^1, 1 + f > 0$. See Fig.1 for the f.

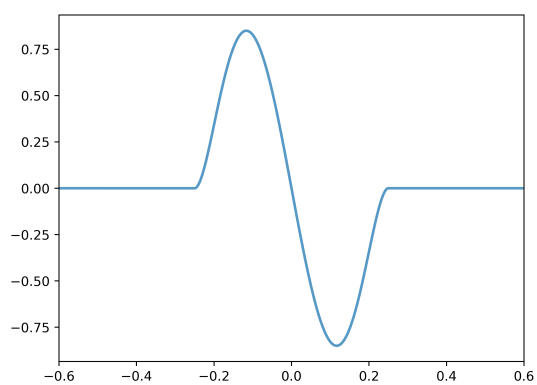

Figure 1: function f. code in grad_A_div/1d/test.py.

Let

$$A(x) = \frac{1}{1 + \sum_{n \in Z, x_0+1 \leq n \leq x_1 - 1} f(x - n - d_n(\omega))}$$

where the displacements $\{d_n(\omega)\}$ are i.i.d random variables taking values in $[-d_{max}, d_{max}]$ with $d_{max} = 1/5$.

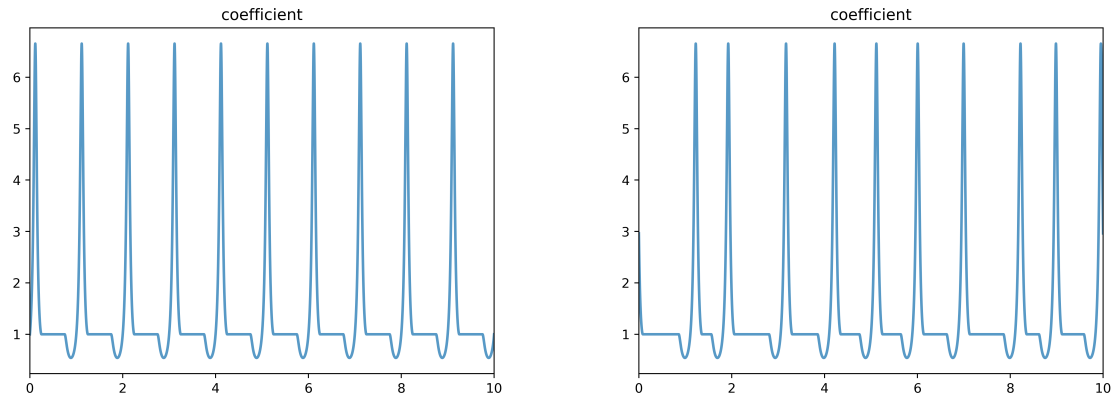

Figure 2: Coefficients on $[0, 10]$. (a). $d_{max} = 0$, i.e., no randomness and 1-periodic. (b) $d_{max} = 0.2$.

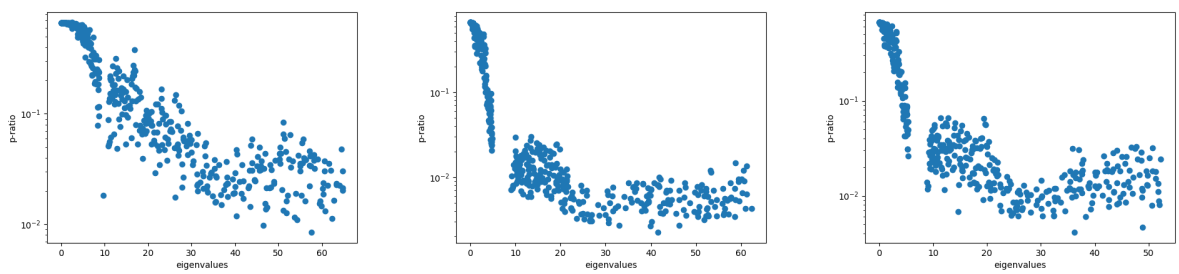

Figure 3: Over $[0, 200]$, Dirichlet boundary, $d_{max} = 0.2$. Participation ratio of first 500 eigenvalues. (a). $-\text{div}(A\nabla u) = \lambda u$. (Results in 1d/Results/000015) (b). $-\text{div}(A\nabla u) = \lambda Au$. (Results in 1d/Results/000016) (c). $-\text{div}(\nabla u) = \lambda Au$. (Results in 1d/Results/000017)

2.2 2d random displacement model

Consider solving

$$-\text{div}(A\nabla u) = \lambda u$$

on a closed square $[0, L] \times [0, L]$. Here,

$$A(x) = \frac{1}{1 + \sum_{n_1, n_2 \in Z} f(x - (n_1, n_2) - d_n(\omega))}$$

where $\text{supp}(f) \subset B_s(0), f \in L^1, 1 + f > 0, f \neq 0$ and the displacements $\{d_n(\omega)\}$ are i.i.d random variables taking values in $B_{d_{max}}(0)$ with $d_{max} + s < 1/2$. Let $s = 1/4$ and define

$$f(x) = 20 \max\{(1 - |x|^2/s^2)^3, 0\}(3|x|^2 + 1) = \begin{cases} 20(1 - |x|^2/s^2)^3(3|x|^2 + 1), & x \in B_s(0) \\ 0, & x \notin B_s(0) \end{cases}$$

Then $\text{supp}(f) \subset B_s(0), 1 + f > 0$.

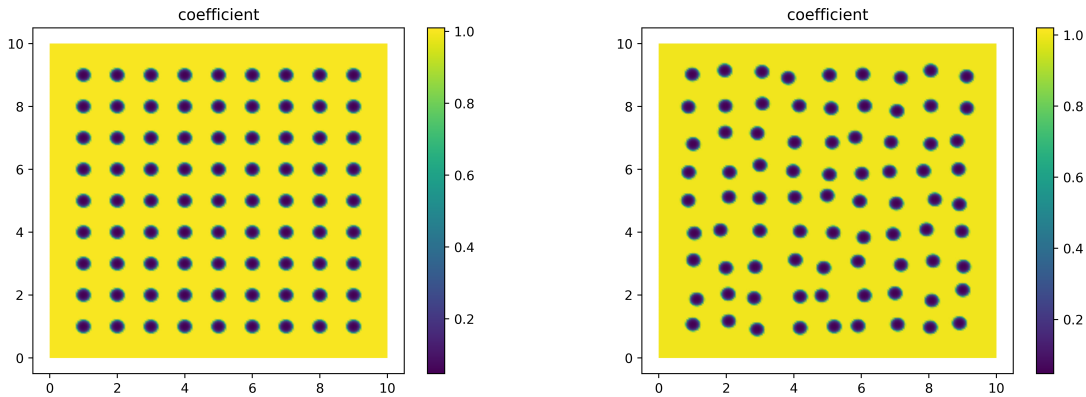

Figure 4: Coefficients on $[0, 10] \times [0, 10]$. (a). $d_{max} = 0$, i.e., no randomness. (b) $d_{max} = 0.2$.

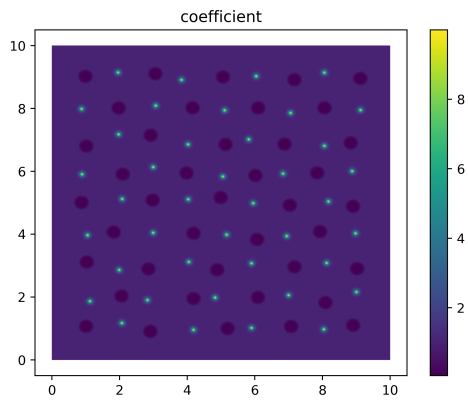

Figure 5: Coefficients on $[0, 10] \times [0, 10]$. another choice of f. $d_{max} = 0.2$.

References

[1] DUERINCKX, MITIA and GLORIA, ANTOINE. Large-scale dispersive estimates for acoustic operators: homogenization meets localization. *Preprint* (2023)

[2] Bordenave, Charles, Arnab Sen, and Balint Virag. Mean quantum percolation. *Journal of the European Mathematical Society* 19.12 (2017): 3679-3707.

[3] Kuchment, Peter. An overview of periodic elliptic operators. *Bulletin of the American Mathematical Society* 53.3 (2016): 343-414.

[4] Sims, Robert, and GÃŒnter Stolz. Localization in One Dimensional Random Media: A Scattering Theoretic Approach *Communications in Mathematical Physics* 213 (2000): 575-597.

[5] Figotin, Alexander, and Abel Klein Localization of classical waves I: Acoustic waves. *Communications in Mathematical Physics* 180.2 (1996): 439-482.

[6] Kuchment, Peter An overview of periodic elliptic operators *Bulletin of the American Mathematical Society* no.3 (2016): 343-414.

[7] Filonov, N Second-order elliptic equation of divergence form having a compactly supported solution *Journal of Mathematical Sciences* 106 (2001): 3078–3086.

[8] Armstrong, Scott and Kuusi, Tuomo and Smart, Charles Large-Scale Analyticity and Unique Continuation for Periodic Elliptic Equations *Communications on Pure and Applied Mathematics* 76 (2023): 73–113